

THOMAS LENZ

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EDUCATION

Iowa State University 2024-2029
Ph.D., Industrial and Manufacturing Systems Engineering and Human Computer Interaction
Human factors research focus
Dr. Jundi Liu, advisor

University of Minnesota-Twin Cities 2020-2024
Bachelor of Industrial and Systems Engineering, Minor in Computer Science
Dr. Nichole Morris, advisor

RESEARCH INTERESTS

- Evaluation of human-vehicle interfaces using physiological and behavioral signals from simulator experiments
- Management of operator workload and situation awareness in autonomous vehicle teleoperation queues
- Design and evaluation of chatbots that encourage appropriate trust and acceptance in users

EMPLOYMENT HISTORY

Iowa State University, Ames, IA
Graduate Research Assistant - HIA Lab *Fall 2024-Present*

- Design and evaluate human-vehicle interfaces in simulated automated driving environments to assess operator workload, trust, and situational awareness
- Collecting data from 32 participants on a UE4 driving simulator to assess display compellingness during partially-automated driving
- Developing an open-source CARLA- and Python API-based simulator for assessing situation awareness during remote assistance of autonomous vehicles
- Mentor undergraduate researchers on experiment preparation, simulator development, and data analysis

Project Mentor - NSF Research Experience for Undergraduates *Summer 2025*

- Mentored a group of three undergraduate students in a summerlong research project
- Developed a CARLA-based simulator to test the effects of queue design on automated vehicle remote operator performance
- Helped students formulate a research question, develop an experiment plan, build a simulator, and test experiment subjects

- Resulted in production of a prototype teleoperation simulator, a pilot study, a draft paper, and a poster presentation

University of Minnesota, Minneapolis, MN

Undergraduate Research Assistant - HumanFIRST Lab

Fall 2023-Fall 2024

- Collected and analyzed data on pedestrian crossing behavior and driver yielding behavior
- Evaluated efficacy of environmental interventions on pedestrian safety

Jr. Engineer Trainee - St. Anthony Falls Laboratory

Fall 2021-Fall 2023

- Assisted professors and graduate students in experimental setup
- Fabricated metal, plastic, and wood apparatuses

MVE, New Prague, MN

Product Engineering Intern

Summer 2023

- Developed testing protocol to assess dewar durability
- Designed experiments to minimize cost of testing while maximizing statistical power
- Evaluated usability of dewars

PUBLICATIONS

1. **Lenz, T.**, EJ Bahng, S Park, J Liu, & B Weng (2026). A Survey of Future Teachers Suggests Cautious Openness to VR and Humanoid Robot Use. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. <https://doi.org/10.1177/10711813261475235>
2. **Lenz, T.**, Brackey, R., & Liu, J. (2025). The effects of citations and confirmation bias on trust in chatbots. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. <https://doi.org/10.1177/10711813251357884>
3. Craig, C., Mabry, M., Drahos, B., Kessler, W., **Lenz, T.**, Olander, K., Morris, N. (2024). An assessment of two pedestrian safety inventories. *Advances in Human Factors of Transportation*. AHFE International Conference. <http://doi.org/10.54941/ahfe1005242>

OTHER SCHOLARLY ACTIVITIES

Conference Presentations

HFES ASPIRE Poster Presentation	<i>Fall 2026</i>
IISE Annual Conference	<i>Spring 2026</i>
IMSE Spring Poster Symposium Presentation	<i>Spring 2026</i>
HFES ASPIRE Poster Presentation	<i>Fall 2025</i>
IMSE Spring Poster Symposium Presentation	<i>Spring 2025</i>
HFES Midwest Regional Meeting Lightning Talk	<i>Spring 2025</i>

Guest Lectures

IE 2710 Applied Ergonomics and Work Design: Cognitive Ergonomics Sampler	<i>Spring 2026</i>
IMSE Departmental Recruitment and Open Houses	<i>Spring 2025-Present</i>

HFES Student Chapter Trivia Host		<i>Fall 2025</i>
HCI 5790X Methods for Interdisciplinary Research: "My Research Journey"		<i>Fall 2025</i>
Professional Society Memberships		
President, HFES Student Chapter at Iowa State University		<i>Spring 2026-Present</i>
Vice President, HFES Student Chapter at Iowa State University		<i>Fall 2025-Spring 2026</i>
Member, Institute of Industrial and Systems Engineers (IISE)		<i>Spring 2026-Present</i>
Member, Human Factors and Ergonomics Society (HFES)		<i>Fall 2024-Present</i>
Member, Human Computer Interaction Student Group		<i>Fall 2025-Present</i>
SKILLS		
Methods: User evaluation, survey design, statistical modeling, design of experiments, human subjects research, machine learning		
Programming and Software: Python, R, Excel, MATLAB, Simio, AMPL, Qualtrics, Generative AI		
Simulation Development: Unreal Engine, Unity, CARLA, VR, eye tracking, physiological signal collection, CAD		
SERVICE		
President, HFES Student Chapter		<i>Spring 2026-present</i>
Vice President, HFES Student Chapter		<i>Fall 2025-Spring 2026</i>
Graduate and Professional Student Senate Representative		<i>Fall 2025-Present</i>
Mentored ten undergraduate research assistants		<i>Fall 2024-Present</i>
AWARDS		
Harold and Shirley Reihman Graduate Scholar	Iowa State University	<i>2024-Present</i>